The rank spectrum of the projective symplectic group in dimension four

Human Compression project: F001 principal assembly

13 October 2026

This paper assembles the complete principal assertion of Brooksbank's fixed preprint [1, Theorem 1.1]. The even-characteristic argument is acquired from the reviewed L031 and L037 recoveries. The odd existence theorem, exceptional-field certificate and upper bound for both involution types have separately reviewed complete proofs. The argument below identifies the precise components required for each field and records the auxiliary assertion that must be corrected.

For a finite group G, let $\mathrm{rk}(G)$ be the set of lengths of its string C-group markings. A marking consists of generating involutions with nonconsecutive generators commuting and with the intersection property for every two subsets of the generating set.

Theorem 1. *For every prime power q,*

$$\mathrm{rk}(\mathrm{PSp}_4(q)) = \begin{cases} \varnothing, & q = 3, \\ \{3,4,5\}, & q \neq 3. \end{cases}$$

Proof. We retain the classical identification $\mathrm{PSp}_4(q) \cong \Omega_5(q)$ in odd characteristic and $\mathrm{PSp}_4(q) = \mathrm{Sp}_4(q)$ in characteristic two. These are also the identifications used in [1, §1]. The group is nonabelian and simple for $q \geq 3$; the case $q = 2$ is treated below without a simplicity assumption.

Suppose first that $q \geq 4$ is even. Theorem 5 of the acquired *Endpoint variation in orthogonal symmetry markings* gives a rank-five string C-marking of $\mathrm{Sp}_4(q)$ by taking its odd-dimensional construction with $2m + 1 = 5$. Every adjacent product has odd order. The acquired rank-contraction theorem and its iterated odd-tail corollary from L037 therefore give ranks four and three for the same group. For the upper bound, a marking of this simple group must be connected. The acquired *A centralizer bound for symplectic strings in dimension four*, Theorem 3, bounds every connected involutory string in characteristic two by five. Thus no rank greater than five occurs.

For $q = 2$, the acquired *Three binary exceptions to a symmetry exclusion*, Lemma 1 and Theorem 2, identifies $\mathrm{Sp}_4(2)$ with the faithful coordinate S_6 acting on the quotient of the even-weight six-coordinate space by its constant line. Its adjacent transpositions form a rank-five string C-marking and all adjacent products have order three. The same two acquired contractions give ranks four and three. Every C-marking is an independent generating family; the retained Whiston bound for S_6 limits its length to five. This argument applies also to disconnected markings, so no simplicity assumption has entered at this boundary.

For every odd $q \geq 5$, the acquired *A free endpoint for odd orthogonal strings in dimension five* proves the existence of ranks three, four and five. It combines a uniform construction for $q \geq 33$, exact parameter rows for $13 \leq q \leq 31$, and exact marked groups for $q = 5, 7, 9, 11$. The finite rows cover every odd prime power below 33, with the case $q = 3$ kept separate. The upper bound is Theorem 4 of *Mixed involutions in orthogonal strings of dimension five*: it treats both nontrivial involution types in the full $\Omega_5(q)$. The previously acquired negative-reflection bound is one of its complete dependencies, not a substitute for the mixed case.

For $q = 3$, the complete finite extension certificate in the odd component proves that $\mathrm{PSp}_4(3)$ has no marking of any rank. Its explicit forty-point projective matrix group has order 25920; the exhaustive prefix search has counts $2, 13, 121, 20, 2$ in lengths one through five and no longer prefix. None generates the group. The accompanying proof establishes complete orbit coverage, valid prefix rejection and termination independently of those observed counts. Simplicity excludes disconnected markings. We acquire that entire proof, the exact programs and their reviewed execution records; we do not infer the result merely from the absence of rank three.

Finally, rank one would give a cyclic group of order two, and a group generated by two involutions is dihedral. None of the groups above is cyclic of order two or dihedral: for $q \geq 3$ this follows from nonabelian simplicity, and for $q = 2$ from S_6. This excludes the remaining ranks and proves the stated spectrum. □

Principal result and auxiliary correction. Theorem 1 recovers the fixed source's principal rank spectrum. The source's Lemma 4.1 asserts a stronger bound for arbitrary subgroups of a five-dimensional orthogonal group; that literal assertion is false. The separate reviewed packet gives the deleted-permutation S_7 marking of rank six over $\mathbf{F}_7$, in both $O_5(7)$ and $SO_5(7)$. Our upper bounds concern the full symplectic or Ω_5 group and do not use that auxiliary lemma. The correction and the recovery of the principal result are therefore distinct claims.

The present construction also replaces the source's rough scalar-density argument by a uniform parameter bound and exact finite boundary proofs. It does not assert that every scalar in the source's displayed ansatz works, or reproduce every marked representation discussed there. In particular the separate four-matrix illustration in [1, §5.3] is not needed for the rank-spectrum contract. The higher-dimensional suggestion in [1, §5.2] remains outside this recovery; no open question is declared resolved by the present assembly.

Attribution and acquired material. The spectrum is Brooksbank's theorem. Symmetry constructions and spinor-norm passage originate in that work and in Brooksbank–Ferrara–Leemans [2]; the contraction is Brooksbank–Leemans [3]. Our replacements use support intersections, explicit parameter control, the full mixed-involution upper bound and the exact exceptional-field certificate. Complete acquired component manuscripts and their required finite records are listed in the accompanying dependency and claim-route files. Classical structure, subgroup and reflection classification inputs remain expressly retained with their external proof costs undetermined. No numerical compression ratio, new historical priority, Lean verification or conjecture credit is asserted here.

References

[1] P. A. Brooksbank, *On the ranks of string C-group representations for symplectic and orthogonal groups*, arXiv:1912.05512v1, 11 December 2019, supplied eleven-page edition.

[2] P. A. Brooksbank, J. T. Ferrara and D. Leemans, *Orthogonal groups in characteristic 2 acting on polytopes of high rank*, arXiv:1709.02219v2, 10 March 2018, supplied twelve-page edition.

[3] P. A. Brooksbank and D. Leemans, *Rank reduction of string C-group representations*, arXiv:1812.01055v2, 3 May 2019; Proceedings of the American Mathematical Society **147** (2019), 5421–5426.

[4] J. Whiston, *Maximal independent generating sets of the symmetric group*, Journal of Algebra **232** (2000), 255–268.